actib

precision you can feel



Arancha Ramirez

learning through vibrations

Training with haptic feedback for enhanced skill development

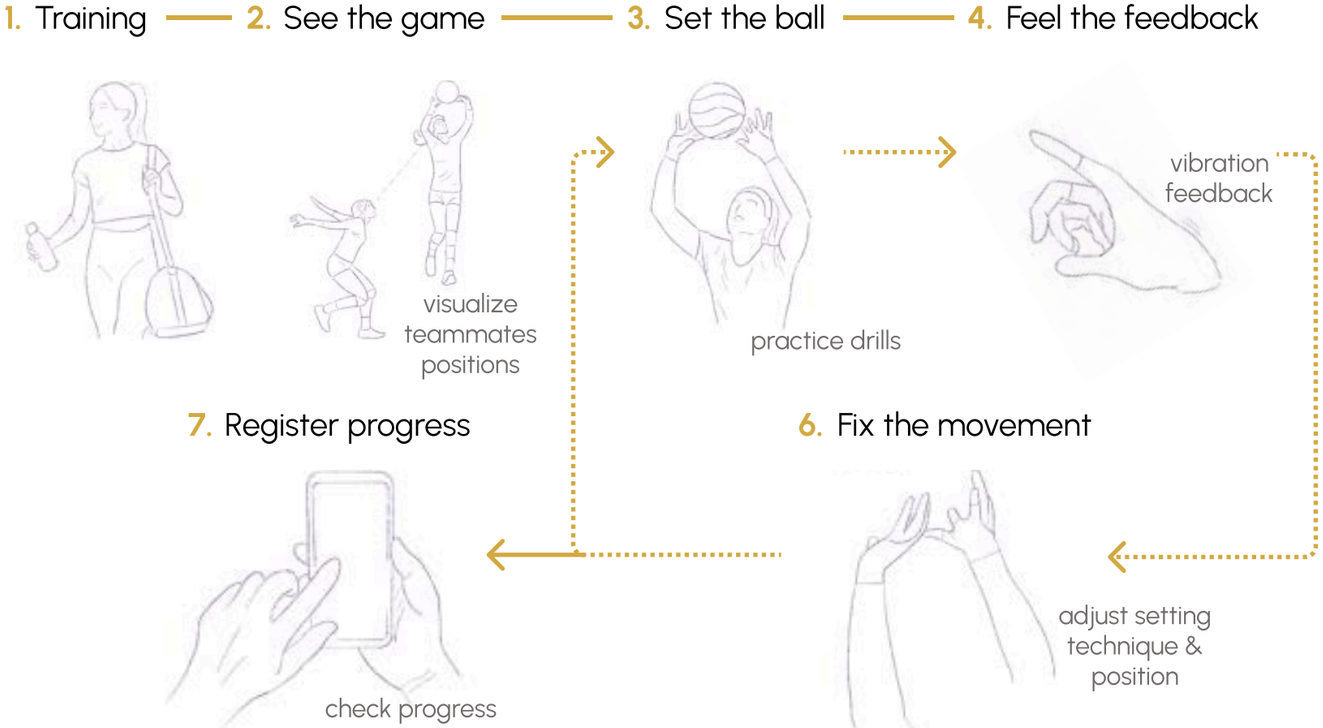
Setters in volleyball require immense concentration during plays, especially in fast-paced game situations. They must constantly assess teammates' positions, anticipate movements, and make split-second decisions to execute precise sets. Their hand positioning and touch must be instinctively accurate, as even minor miscalculations can disrupt an entire offensive play.

To refine their skills, they need instant, real-time feedback during practice and drill sessions, allowing them to make adjustments on the fly without breaking focus.

understanding fast play

Good training tools provide immediate, intuitive feedback to reinforce correct technique. In volleyball, precision in hand positioning and force application is critical.

→ Haptic feedback translates real-time data into tactile cues, allowing athletes to adjust their movements instinctively during drills.



the problem

Traditional training methods rely on coaches' verbal feedback or video analysis, which delays correction and limits real-time skill improvement. Athletes struggle to self-correct during practice, leading to slower progress.

HMW design a wearable that helps volleyball players improve their setting technique by providing real-time feedback on accuracy and hand positioning?

key insights

- Athletes benefit from instant feedback rather than post-session analysis during drills.
- Overloading with visual/audio cues can be distracting even in controlled practice environments.
- Vibrational cues allow for intuitive, non-intrusive correction in training sessions, building confidence before game situations.

design opportunity

- Integrate precise haptic feedback into wearable technology for instant technique correction in training.
- Ensure customizable feedback intensity for different drill types and skill levels.
- Develop a system that trains hand positioning and force application in structured practice scenarios.

design requirements

- **Simplicity:** Minimal distraction while delivering effective feedback during training sessions.
- **Function:** Must be responsive and accurate in detecting errors in a practice setting.
- **Practicality:** Comfortable and usable during drills, ensuring seamless integration into training routines.

the design

Actib is a wearable training device designed to refine setting accuracy and technique for volleyball players. By integrating real-time haptic feedback, actib provides setters with instant corrections during drills, ensuring optimal hand positioning, force application and wrist movement.

Designed for repetition intensive training sessions, actib features moisture-wicking materials to **reduce sweat buildup**, ensuring consistent grip and preventing slips during setting drills. Its finger-less design ensures direct contact with the ball's texture, enhancing precision, while targeted resistance zones help **strengthen fingers** and the structured fit reinforce **proper technique** for controlled, consistent sets. And its haptic feedback module, positioned on the upper wrist but below the joint, delivers **real-time tactile corrections** without restricting movement, ensuring unobtrusive yet effective training feedback.

With its ergonomic form and responsive feedback technology, Actib enables players to develop **precision, control,** and **consistency** before stepping into real-game scenarios.

technical details

exposed fingertips

The exposed fingertip design provides direct contact with the ball, improving tactile feedback and precision. This allows players to feel the ball's texture for better control and subtle technique adjustments, making it ideal for high-speed drills and improving overall performance.



motion sensors

Including advanced motion sensors enables actib to track the player's movements with precision. The data syncs with a mobile app or training software, providing detailed analytics on hand trajectory, wrist angle, and finger positioning, helping players track progress and make data-driven improvements.

structured fit

Actib's structured fit is designed for comfort and performance. The ergonomic design adapts to the player's hand and wrist with flexible, moisture-wicking fabrics, ensuring unrestricted movement while reinforcing proper technique. This tailored fit supports finger control and wrist positioning, optimizing form during repetitive setting drills.

customizable fit

Featuring adjustable straps around the wrist and hand for a personalized, secure fit across various hand sizes. Made from stretchable, breathable material, the straps ensure comfort, moisture management, and a firm hold during intense training sessions.

feedback mechanism

Actib's haptic feedback module delivers real-time corrections via gentle vibrations on the upper wrist. This unobtrusive feedback helps players adjust hand positioning, force, and wrist movement instantly, ensuring consistent and controlled sets without interfering with natural motion.

aesthetic & form details

c.m.f selection

To enhance both functionality and aesthetics, actib utilizes a combination of durable and performance-driven materials:



Silicone Coating
(High-Wear Resistance)

Provides enhanced durability in high-contact areas, offering abrasion resistance while maintaining flexibility for comfort and prolonged use.

Recycled ABS
(Matte or Semi-Gloss Finish)

Provides durability while maintaining a lightweight design, ideal for active use.



Poliamide
(Flexible & Impact-Resistant)

Offers flexibility and impact resistance, ensuring a secure fit while withstanding intense training conditions.

removable electronics

In assessing actib's desirability, an essential form consideration was ensuring the wearable could be sanitized, to eliminate the sweat build-up caused by the nature of the exercise.

Actib's modular design ensures the core electronic components can be detached, allowing for easy washing, maintenance, and repairs.

This ensures the wearable remains functional and long-lasting without compromising hygiene or durability.

storage case

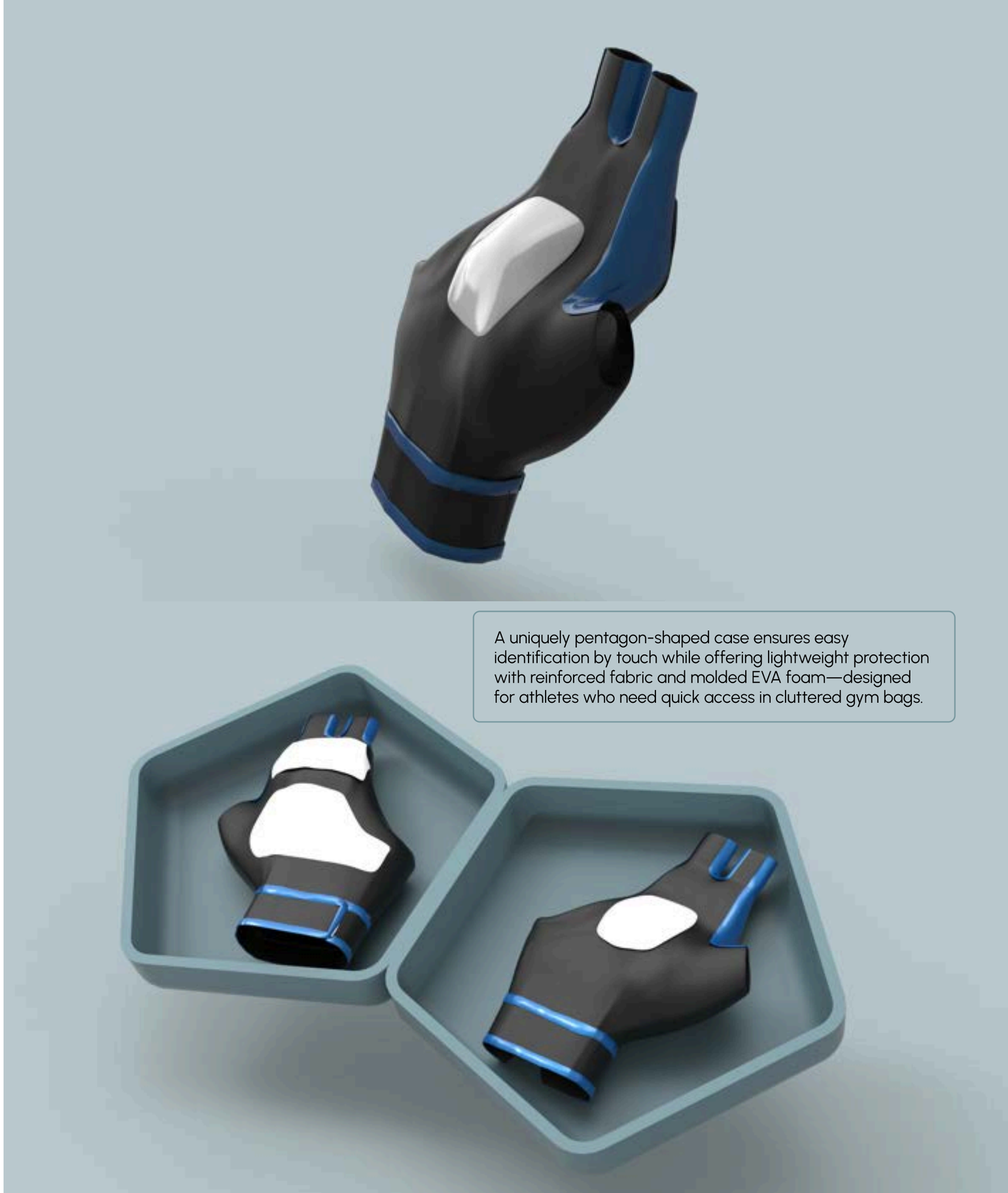


Anna, Imperial Women's Setter

"Between my knee pads, shoes, sleeves and stretch bands my gym bag is incredibly cluttered... I can never find anything. Could the glove come in a unique case so they're easy to identify by touch alone?"

Through interviews with players, a vital form consideration was ensuring compactness and easy transport.

Designed for athletes on the go, actib fits into a small, travel-friendly case that seamlessly fits in gym bags. This feature isn't just about protection - it's about enabling intuitive identification and aligning with user's needs.



A uniquely pentagon-shaped case ensures easy identification by touch while offering lightweight protection with reinforced fabric and molded EVA foam—designed for athletes who need quick access in cluttered gym bags.

initial concept exploration

understanding setters needs

To analyze the setter's role, multiple matches were analyzed, focusing on how hand positioning and wrist movement influenced the accuracy and effectiveness of sets.





It was noted how competitive setters adjusted their wrist angle, finger spread, and force application to deliver precise sets, ensuring optimal ball trajectory for hitters.

Variations in hand technique affected the speed, spin, and height of the set, directly impacting the success of attacks.







Additionally, we examined how setters adapted under pressure, adjusting for quick transitions, off-balance plays, and defensive disruptions.

From these observations, key challenges were identified :

- inconsistent wrist stability led to misdirected sets
- excessive force altered ball control
- improper finger placement reduced accuracy
- lack of real-time feedback, making it difficult to self-correct errors during training

These insights shaped the design, emphasizing haptic feedback for immediate corrections, structured support for wrist stability, and an open-fingertip design for better ball feel—ensuring setters develop precision, consistency, and control in their technique.

concept sketches

Early hand-drawn sketches helped visualize different wearable forms, possible sensor placement, and material layouts to ensure functionality and comfort.

Four types of designs where ideated:

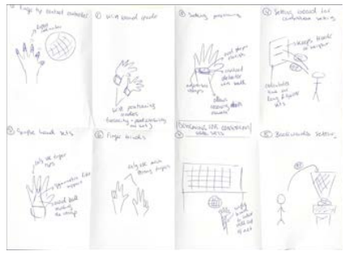
- precision tracking
- correct hand technique monitoring
- hand support during setting
- feedback through ball interaction



brainstorming

Rapid sketching was used to explore potential form factors aimed at improving setters' precision. Each of the 15 initial concepts was developed in 30-second intervals, allowing for quick idea generation without over-analyzing details.

The designs focused on various feedback mechanisms and support structures.



Evaluating the sketches revealed that most set-related issues stemmed from hand positioning and wrist movement. With few market solutions, hand-focused design was prioritized.

morphological analysis

Functions and features were explored through morphological analysis.

Accuracy Tracking	score board	collection net	adjustable hoop	
Hand Support	wrist support	knuckles aid	fingertip anti slip	forearm cushion
Haptic Feedback	wrist vibration	index & thumb control	palm control	3 finger sleeve
External Tracking	ball sleeve	camera tracking	smart watch positioning	

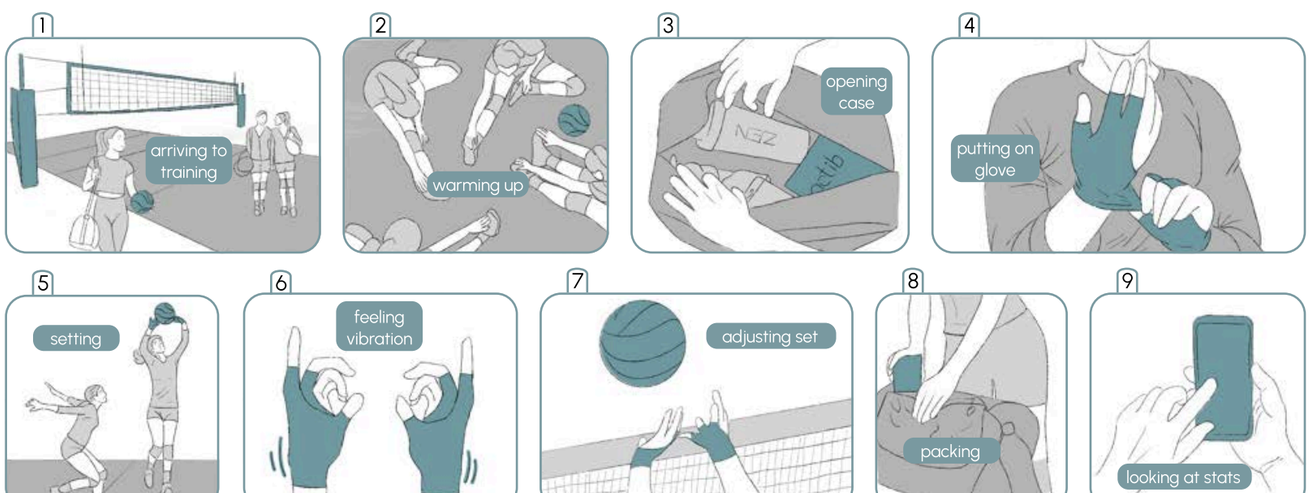
form exploration

Sleek, minimal, dynamic design that doesn't stand out too much, for easier integration.



storyboard

A sequence of illustrated scenarios depicting how players might interact with actib device during training was used to visually map the user's steps and highlights.

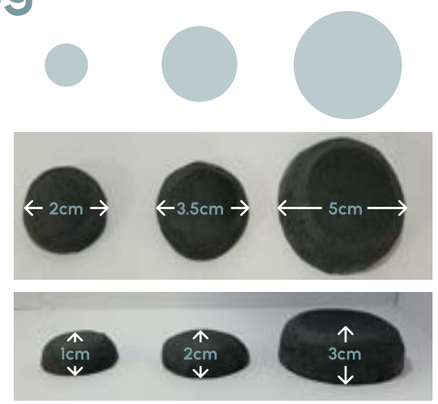


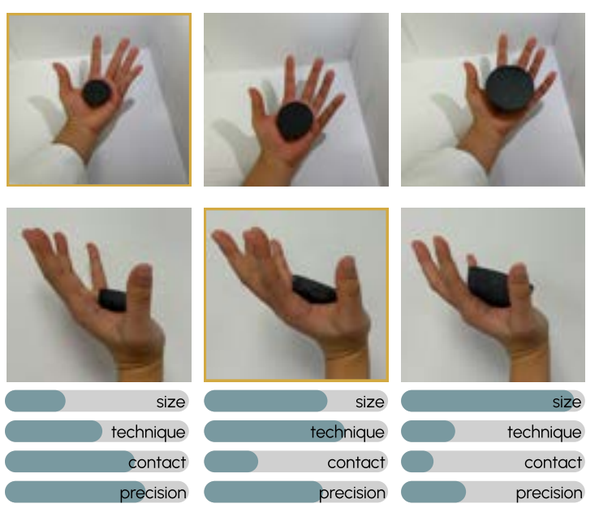
lo-fi prototyping

palm support size & positioning

Three different palm support sizes and curvatures were prototyped, ranging from 2 cm to 5 cm, to evaluate their fit with the natural hand shape and their impact on setting technique. Each prototype varied in thickness and contour, aiming to find the right balance between support and comfort without restricting movement.

To test their effectiveness, the prototypes were placed on the palm and compared across various setting motions. Players performed controlled sets, evaluating differences in technique, ball contact, and precision. Observations focused on how each size influenced hand positioning and overall ball control.





The smallest height and curvature were selected for their unobtrusive design and natural fit, ensuring minimal interference with hand movement.

However, testing revealed that the medium-sized surface area provided better hand positioning during sets.

This insight was incorporated into the next design iteration by combining the 1cm height and 2cm curvature with a medium-sized support surface to optimize both comfort and performance.

finger coverage



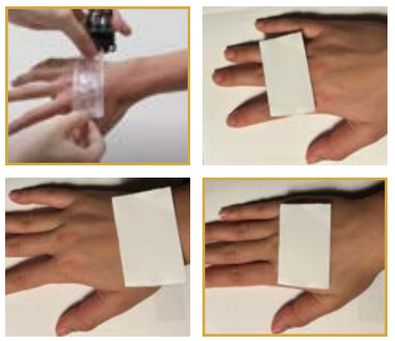
Finger contact with the ball was analyzed using the small prototyped design and compared to elite setters' techniques to identify key points of contact.

While setting motions varied in the number of fingers used, the thumb, pointer, and middle fingers consistently maintained ball contact during setting, so the design prioritized their support while allowing natural movement for better precision and control.

Building on these insights, the design was developed with coverage focused on the 3 selected fingers. Testing revealed that greater ball contact improved set control, so, to ensure consistency, the glove was designed to extend just below the middle knuckles.

motion sensor positioning

Strain gauge sensors were integrated into the motion tracking system to measure the subtle deformations in the hand's structure during movement. They work by detecting changes in resistance as the material deforms, allowing for precise tracking of the force and movement applied to specific areas of the hand.



To identify their optimal position, they were placed on the back of the hand, with particular focus on areas around the metacarpal bones, just above the wrist and below the knuckles, where the hand experiences noticeable changes in shape and strain during motion.

The sensors were most effective when placed on the upper back of the hand, near the base of the finger, where they could accurately measure wrist flexion and extension without restricting motion. while also capturing small movements and finger positioning.

haptic feedback positioning

Three different band widths were tested to analyze their impact on comfort, range of motion, and unobtrusiveness. Narrow bands (1-2 cm) provided minimal interference but lacked stability, while wider bands (3-4 cm) ensured a secure fit but restricted wrist movement.



A balanced 2 cm band was found to be optimal, offering both secure placement and freedom of motion while effectively transmitting haptic feedback.

For the ideal positioning, the haptic module was tested on different areas of the hand and wrist.



Placing it on the palm interfered with ball contact, while a placement directly on the wrist joint restricted movement. The most effective location was found to be just above the wrist but below the joint on the back of the hand. This position allowed the vibrations to be easily felt without obstructing natural wrist motion, ensuring the feedback remained noticeable yet unobtrusive during play.

final device architecture

The design incorporates a 1cm height, 2cm curvature, and a medium-sized support surface for optimal comfort and performance.

Coverage focuses on three selected fingers, with the glove extending just below the middle knuckles to enhance set control.

Sensors are placed on the upper back of the hand near the base of the fingers to track wrist and finger movements accurately.

The optimal feedback location is just above the wrist but below the joint, ensuring vibrations are felt without restricting motion.

design exploration

finger articulation

To enhance flexibility, the glove incorporates 1.5mm deep flex grooves at key finger joints. These articulation points allow for greater range of motion compared to standard glove designs, ensuring setters can maintain precision without restriction.

overlays

Durability is enhanced with 0.5mm synthetic overlays placed on the thumb, index, and middle fingers—areas that experience the most friction. This reinforcement extends glove lifespan, preventing premature wear without compromising flexibility.



mesh zones

A combination of perforated mesh panels (1mm micro-perforations) on the back of the hand and a moisture-wicking lining prevents sweat buildup, keeping hands dry and improving grip consistency during training.

snug fit

The glove features a compression-fit design with a 2mm stretchable Lycra layer, ensuring a secure yet flexible fit. This allows natural hand movement while preventing excess material from interfering with ball control. The streamlined pattern reduces the number of fabric pieces required, simplifying the cutting and assembly process during manufacturing.

wrist support

A 4cm-wide elastic wrist strap with velcro adjustments provides personalized compression, reducing strain on the wrist. This feature helps absorb impact forces, lowering fatigue during repetitive setting drills.

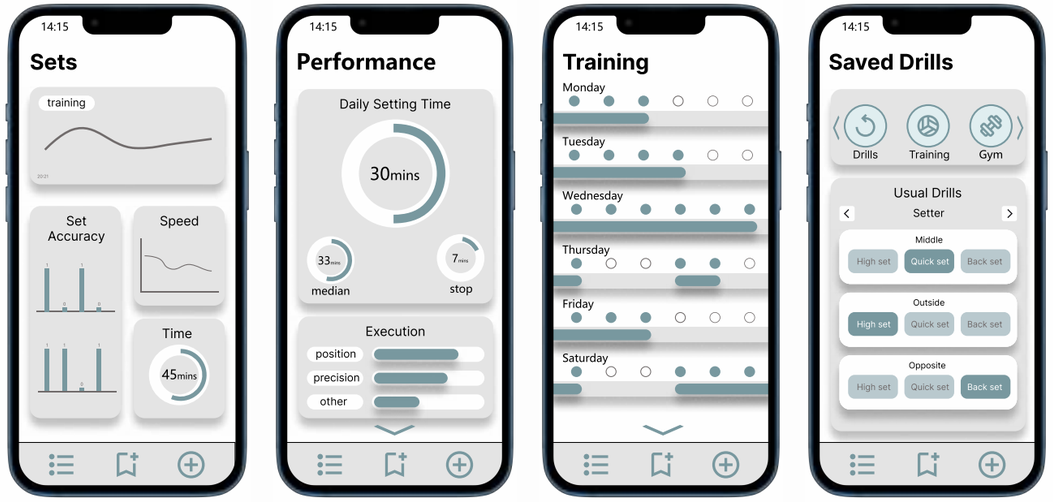
minimal seams

The palm is constructed using a single-panel seamless design, eliminating pressure points that could cause discomfort. With flatlock stitching (only 0.5mm thick) at key joins, the glove maintains a smooth interior, enhancing tactile feedback and ensuring uninterrupted ball contact.

user interaction

the app

An application comes with the actib gloves. The actib app is connected to the gloves and provides monitoring and feedback. It guides the players through the selected drills, following the selected preferences and providing haptic feedback after each set. It also records training times, precision and hand position to improve technique and provide tailored feedback.



customizable feedback

Actib features interchangeable feedback modules, allowing players to choose from different types of haptic feedback based on their preference or training focus.



Anna, Imperial Women's Setter

"Being able to adjust the feedback intensity would be very useful. I'd be able to use stronger vibrations when I'm focusing on fixing specific mistakes and softer ones when I want to fine-tune my hand positioning without too much distraction."



Alix, Imperial Women's Setter

"Customization would help me tailor each training session. When I'm drilling basic mechanics, I'd keep the feedback light, but when working on precision and consistency, I'd increase the intensity to ensure every movement is correct."

Their feedback was taken into account and implemented by allowing players to customize feedback intensity by choosing between modules that deliver stronger vibrations or softer feedback.

This modular approach allows them to fine-tune their training experience and select the level of guidance that best fits their needs.

compatibility



Catalina, Imperial Women's Middle Player

"I already use a fitness tracker—can Actib sync with it to track my training and recovery together?"

To enhance its versatility, actib can sync with popular fitness trackers like Fitbit and Garmin, enabling players to monitor additional metrics and training load for a comprehensive view of their sessions.



